

We have found major changes in a plant-pollinator network over the past 120 years. This is partly explained by the nonrandom extirpation of bee species that are expected to be the most vulnerable to land-use and climate change, such as rare and specialized species, species occupying higher trophic levels, and cavity-nesting species. We found large changes in phenology of both forbs and pollinators and the potential for interaction mismatches, and these phenological changes can explain some of the species and interaction losses observed in this system. Our more optimistic finding was that plant-pollinator interaction networks were quite flexible in the face of strong phenological change and bee species extirpations, with many extant species gaining interactions through time. However, the redundancy in network structure has been reduced, interaction strengths have weakened, and the quantity and quality of pollinator service has declined through time. Further interaction mismatches and reductions in population sizes are likely to have substantial negative consequences for this crucial ecosystem service.

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Supplementary Materials

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Materials and Methods

Supplementary Text

Figs. S1 to S10

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Roots and Associated Fungi Drive Long-Term Carbon Sequestration in Boreal Forest

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Boreal forest soils function as a terrestrial net sink in the global carbon cycle. The prevailing dogma has focused on aboveground plant litter as a principal source of soil organic matter. Using ¹⁴C bomb-carbon modeling, we show that 50 to 70% of stored carbon in a chronosequence of boreal forested islands derives from roots and root-associated microorganisms. Fungal biomarkers indicate impaired degradation and preservation of fungal residues in late successional forests. Furthermore, 454 pyrosequencing of molecular barcodes, in conjunction with stable isotope analyses, highlights root-associated fungi as important regulators of ecosystem carbon dynamics. Our results suggest an alternative mechanism for the accumulation of organic matter in boreal forests during succession in the long-term absence of disturbance.

Globally, the boreal forest biome covers 11% of the land surface (1) and contains 16% of the carbon (C) stock sequestered in soils (2). Aboveground plant litter quality and decomposition rates have been proposed as the fundamental determinants of long-term soil organic matter accumulation (3–6). However, a large proportion of photosynthetically fixed C is directed belowground to roots and associated microorganisms (7, 8), potentially affecting C sequestration either positively or negatively (9–12). A better mechanistic understanding of how the belowground allocation of C affects

long-term sequestration rates is crucial for predictions of how the currently large C stock in boreal forest soils may respond to altered forest management practices, climate change, elevated CO₂ levels, and other environmental shifts.

Here we present evidence from a fire-driven boreal forest chronosequence that enables the study of soil C sequestration over time scales of centuries to millennia. The system consists of forested islands in two adjacent lakes, Lake Hornavan and Lake Uddjaure (65°55' to 66°09'N; 17°43' to 17°55'E), in northern Sweden. The islands in these lakes were formed after the most recent

glaciation and have since been subjected to similar extrinsic factors. Larger islands, however, burn more frequently because they have a larger area to intercept lightning strikes (6, 13); several large islands have burned in the past century, whereas some small islands have not burned in the past 5000 years. It has previously been shown that as the time since fire increases, soil and total ecosystem C accumulates unabated and linearly (6, 14), leading to humus layers that can exceed 1 m in depth on the smallest islands. This has been attributed to a decline in the quality of aboveground litter inputs and impaired litter decomposition as the chronosequence proceeds (6, 14, 15). We studied organic soil profiles on 30 islands representing three size classes with increasing belowground C stocks (14): 10 large islands (>1.0 ha; on average, 6.2 kg of C m⁻² accumulated belowground; mean time since fire 585 years), 10 medium islands (0.1 to 1.0 ha, 11.2 kg of C m⁻², 2180 years), and 10 small islands (<0.1 ha, 22.5 kg of C m⁻², 3250 years).

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We explored C dynamics across the chronosequence by analyzing bomb ^{14}C (16) to determine the age since fixation of soil C, and then fitted a mathematical model to measurements of C mass and age distribution across vertical organic matter profiles for six representative islands; three large and three small (Fig. 1) (17). The model assumes two sources of C inputs: (i) a series of consecutively deposited cohorts of aboveground plant litter with negligible vertical mixing, and (ii) belowground inputs through root transport and rhizosphere processes. The dynamics of both C sources were estimated by a Bayesian parameterization of the model. The observed distribution of C mass and age was adequately predicted only when root C import was accounted for (Table 1 and figs. S1 and S2). The parameterized model estimates that the proportion of root-derived C accumulated over the past 100 years is larger on small islands (70%) than on large islands (47%), and the larger total C sequestration on small islands during this period can be explained entirely by root-derived inputs (Fig. 1). Differences in organic matter accumulation between islands were primarily determined by processes at the interface between the fragmented litter (F) and humus (H) layers, which corresponded to the zone of highest root density (Fig. 2B) and where the aboveground litter was 10 to 60 years old. The model was run for 100 years, covering almost the entire humus profile of the large islands, but on small islands a major proportion of C is stored in deeper horizons that are older than this. However, the model indicates that below 20 cm depth, root-derived C inputs are low and the C remaining from the horizons above decomposes slowly, as is also supported by ^{14}C depletion in the deeper layers of small islands (Table 1). Thus, root-mediated C input to the upper part of the profile represents a major contribution to the long-term buildup of humus, especially in late successional ecosystems.

Fungi play central roles in boreal forest ecosystems, both as decomposers of organic matter and as root-associated mediators of belowground C transport and respiration. We profiled the relative abundance of major functional groups of fungi through the depth profile of each island by DNA barcoding based on 454 pyro sequencing of the ITS2 region of ribosomal RNA genes (17, 18). These analyses suggest that fungal communities in the uppermost litter layers were dominated by free-living saprotrophs, whereas mycorrhizal and other root-associated fungi dominated at greater depth (Fig. 2A). Thus, root-associated fungi dominate the part of the soil profile where the model indicates the largest difference in C sequestration between the island size classes. At this depth, free-living saprotrophs (mainly molds and yeasts) make a much reduced contribution, suggesting a correspondingly greater role of root-associated fungi in the regulation of organic matter dynamics.

The increase in root-derived C sequestration as the chronosequence proceeds is matched by a

shift in the balance between the production and decomposition of fungal mycelium in the F-H transition zone of the soil profile. We measured the fungal-specific cell membrane lipid ergosterol as a marker for fungal biomass throughout each soil profile. Even though standing fungal biomass, as indicated by total (free plus bound) ergosterol (Fig. 3, A and B) and ITS copy numbers (table S1), was roughly similar on all islands, free ergosterol (characteristic of newly formed mycelia) (19–21) was about 20 times more abundant on large than on small islands, indicating a larger proportion of freshly produced mycelium and thus greater mycelial production. In contrast, bound ergosterol (the proportion of which increases during mycelial senescence) (19, 20) was more abundant on smaller islands, indicating older mycelium with slower biomass turnover.

Furthermore, the fungal cell-wall polysaccharide chitin (Fig. 3C) peaked in the F layer and declined in lower horizons of large islands, but remained at high concentrations at greater depths on the small islands. Chitin persists longer than ergosterol in fungal tissues after death (21), and the high level of chitin on small islands suggests retarded decomposition of fungal cell wall residues. Thus, in spite of supposedly greater mycelial production on the large islands, less mycelial necromass accumulated there than on small islands, suggesting that the large production was counterbalanced by faster decomposition of mycelial remains. Correspondingly, the ^{14}C model indicated faster decomposition of root-derived C on large islands, despite inputs being conservatively constrained to be equal across all islands. Taken together, our results point to impaired

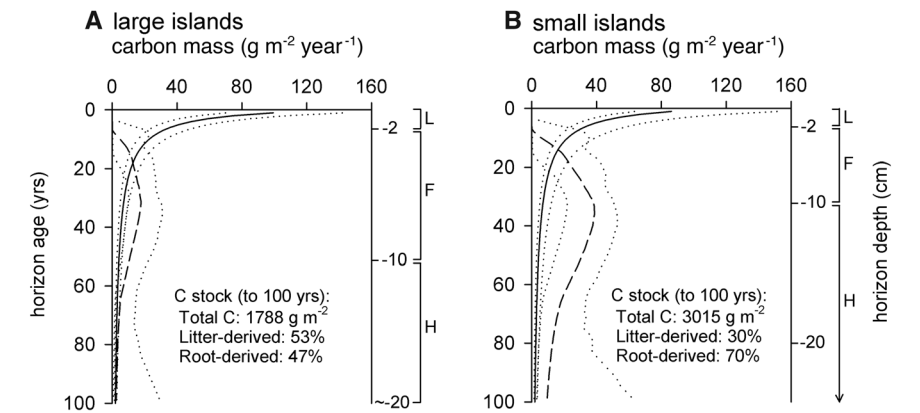


Fig. 1. Carbon dynamics in vertically stratified organic horizons of forested islands. Model estimates of C from aboveground litter (solid lines) and C introduced belowground via root transport (broken lines) are shown. C mass was modeled to a horizon age of 100 years, based on C mass and ^{14}C measurements in profiles from three large (A) and three small (B) islands. Dotted lines show the 95% central credibility intervals around posterior means. The posterior probability that the root-derived fraction is larger on small islands than on large islands is 0.97. Approximate depths are indicated for transitions between the main categories of horizons sampled; L, litter; F, fragmented litter; H, humus.

Table 1. C mass, ^{14}C abundance, and estimated C mean age of sampled organic layers on large and small islands. Means $\pm$ SE, $n = 3$ ($n = 2$ for the deepest layer in both size classes; both values are given).

Layer	Large islands			Small islands		
	C mass (g m ⁻²)	$\Delta^{14}\text{C}$ (‰)	C age (years)	C mass (g m ⁻²)	$\Delta^{14}\text{C}$ (‰)	C age (years)
Litter, on surface	103 $\pm$ 3	70 $\pm$ 3	7 $\pm$ 0	95 $\pm$ 7	62 $\pm$ 1	6 $\pm$ 0
Litter, 0 to 2 cm	185 $\pm$ 17	81 $\pm$ 17	9 $\pm$ 3	180 $\pm$ 11	84 $\pm$ 11	9 $\pm$ 2
Fragmented litter, 2 to 7 cm	454 $\pm$ 91	118 $\pm$ 12	15 $\pm$ 2	539 $\pm$ 63	101 $\pm$ 14	12 $\pm$ 2
Fragmented litter, 7 to 10 cm	493 $\pm$ 76	165 $\pm$ 19	20–39*	494 $\pm$ 77	136 $\pm$ 7	18 $\pm$ 1
Humus, 10 to 16 cm	1170 $\pm$ 90	150 $\pm$ 31	42–51*	1150 $\pm$ 165	191 $\pm$ 14	23–33*
Humus, 16 to 20 cm	1620 $\pm$ 230	21 $\pm$ 16	53 $\pm$ 1	910 $\pm$ 105	195 $\pm$ 17	34–50*
Humus, 20 to 40 cm	3410, 1846	9.4, -14.3	53, 57	6440 $\pm$ 815	14 $\pm$ 15	59 $\pm$ 7
Humus, 40 to 60 cm				7660 $\pm$ 700	-99 $\pm$ 14	780 $\pm$ 120
Humus, 60 to 80 cm				7380 $\pm$ 1760	-194 $\pm$ 9	1670 $\pm$ 90
Humus, 80 to 100 cm				5620, 3650	-285, -267	2628, 2432

*The mean age is within the given interval in samples that include the 1960s peak in bomb ^{14}C .

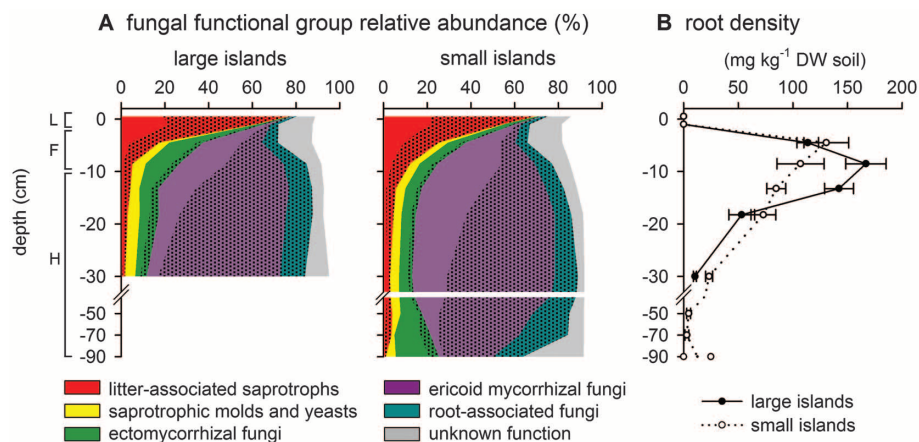


Fig. 2. Depth profiles of the relative abundance of fungal functional groups (percent of amplified ITS sequences) (A) and root (1 to 5 mm in diameter) density (B) on large and small forested islands. The profile corresponds to the organic layers L, F, and H. Functional groups comprise identified species with known function (unshaded) and species putatively assigned to a function (shaded) (17). The data set contains 650,000 sequence reads, and the globally most abundant 583 clusters are analyzed, covering 82 to 95% of the reads in individual sample types. The total ITS copy number was not affected by island size but decreased with depth (with 3×10^9 , 4×10^8 , and 2×10^7 copies g^{-1} of organic matter in L, F, and H layers, respectively) (table S1). The abundances of different functional groups should be compared with caution because of possible differences in ITS copy numbers per unit of biomass. All values are based on means of $n = 10$ islands (except that $n = 2, 5$, and 7 for the lowest horizons) (17).

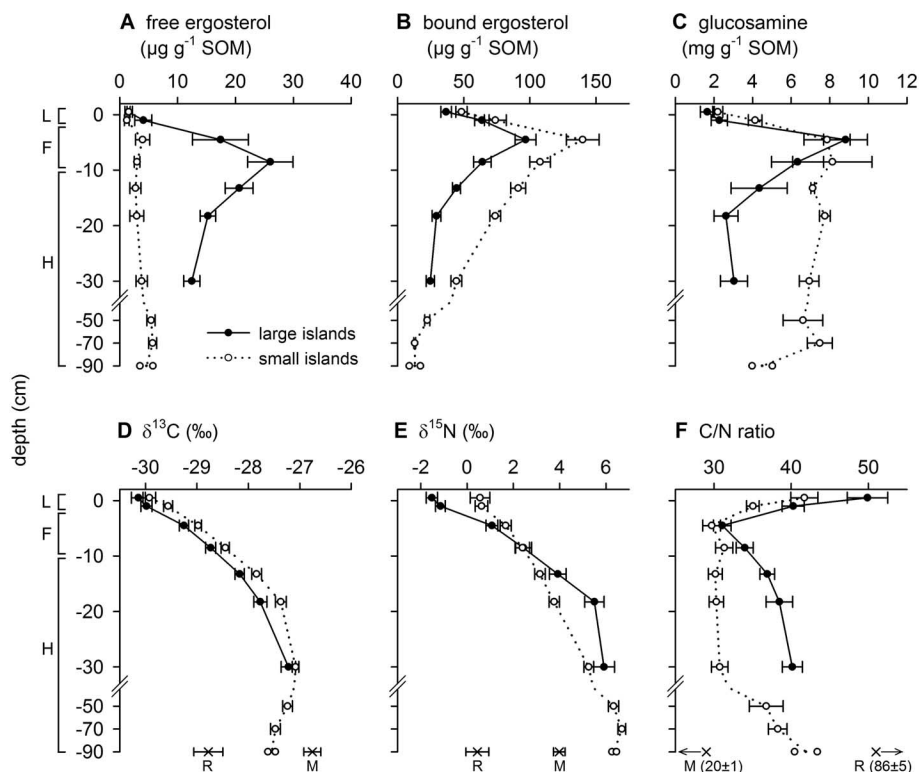


Fig. 3. Depth profiles of fungal biochemical markers (A to C), $\delta^{13}\text{C}$ (D), $\delta^{15}\text{N}$ (E), and the C/N ratio (F) in organic soil profiles of large (solid lines) and small (broken lines) forested islands. All data are means $\pm$ SE, $n = 10$ (except that $n = 3$ for glucosamine and $n = 2, 5$, and 7 for the lowest horizons) (17). Medium-sized islands are not shown but are included in the statistical analyses presented in table S1. In the lower panel, levels measured in roots (R) 1 to 5 mm in diameter and mycorrhizal mycelium (M) sampled at 10 cm depth are given for reference.

decomposition of fungal residues as an important regulator of C accumulation as the chronosequence proceeds.

The often observed increases in ^{13}C and ^{15}N abundances with soil depth have been interpreted as evidence of increasing contributions of enriched microbial components to residual soil organic matter (22–24). In our system, where the dominant plant species form ecto- or ericoid mycorrhizal associations, the $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ signatures in the uppermost organic layers were similar to those of leaves (25), whereas signatures in humus layers were closer to those of rhizosphere mycelium (Fig. 3, D and E, and table S1) and mycorrhizal fungal sporocarps (26). Plant C allocated belowground is relatively enriched in ^{13}C , and this enrichment is further accentuated during C transfer to mycorrhizal fungi (27). However, historic changes in atmospheric $\delta^{13}\text{C}$ (28) may also contribute to the depth gradient (22, 24). Mycorrhizal fungi also have higher $\delta^{15}\text{N}$ signatures than their host plants, because they supply N to their hosts that is ^{15}N -depleted relative to that retained in their own mycelium (29). Thus, the incorporation of isotopically enriched root and fungal remains is likely to be an important mechanism behind the increasing stable isotope signatures with soil depth in this system. This is consistent with the observation that isotopic signatures remain relatively constant in the initial litter decomposition phase and only increase when root-associated fungi dominate C and N dynamics (Figs. 2 and 3) (30, 31).

Previous studies in this (6, 14) and other (3, 4) systems have pointed to the input and quality of aboveground litter as important regulators of C and N sequestration during long-term ecosystem development and succession. Our results show that aboveground plant litter dynamics on its own cannot explain the increasing rate of organic matter accumulation with time since wildfire, and that the dynamics of roots and associated fungi is an important additional factor explaining C accumulation in boreal forests. Although we observe less C accumulation on large islands, it is reasonable to assume that C allocation to roots and associated mycelium is greatest on those islands, especially given their higher root densities (Fig. 2B), free ergosterol levels (Fig. 3A), and net primary productivity (6). This apparent contradiction corroborates recent results (32, 33) showing that increased C input to roots in response to CO_2 enrichment accelerates the turnover of soil organic matter, counteracting C accumulation and enhancing N cycling through the microbial pools. In our system, a similar stimulation of N recycling by large C inputs is supported by the steeper ^{15}N gradient (31) and higher C:N-ratio in the humus of large islands (Fig. 3, E and F) (30). In contrast, the less steep ^{15}N gradient and lower C:N-ratio on smaller islands suggest impaired mycorrhizal N mobilization (31) and accumulation of N in biochemically stabilized fungal remains, consistent with the high levels of bound ergosterol and

chitin on those islands. The consequential reduced N availability to plants leads to progressive nutrient limitation and compositional changes in the vegetation with increasing time since a major disturbance (14, 34). Changes in plant productivity and community composition may, in turn, influence total belowground C allocation and distribution to fungal associates. Together, these feedbacks result in continuing C and N accumulation in the humus layer and decreasing plant production, and this process is only reset by major disturbances, such as wildfire.

Our results elucidate the mechanisms underpinning C sequestration in boreal forests and highlight the importance of root-associated fungi for ecosystem C balance and, ultimately, the global C cycle. We challenge the previous dogma that humus accumulation is regulated primarily by saprotrophic decomposition of aboveground litter, and envisage an alternative process in which organic layers grow from below through the continuous addition of recently fixed C to the organic matter profile in the form of remains from roots and associated mycelium. Environmental changes, such as N fertilization and deposition, forest management, and elevated atmospheric CO₂ concentrations, are therefore likely to greatly affect soil C sequestration through their alteration of rhizosphere processes. These processes are not well described in current models of ecosystem and global C dynamics, and their more explicit inclusion is likely to improve both the mechanistic realism and future predictive power of models.

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Supplementary Materials

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The Biological Underpinnings of Namib Desert Fairy Circles

Norbert Juergens

The sand termite *Pсамmotermes allocerus* generates local ecosystems, so-called fairy circles, through removal of short-lived vegetation that appears after rain, leaving circular barren patches. Because of rapid percolation and lack of evapotranspiration, water is retained within the circles. This process results in the formation of rings of perennial vegetation that facilitate termite survival and locally increase biodiversity. This termite-generated ecosystem persists through prolonged droughts lasting many decades.

Fairy circles (FCs) are large, conspicuous, circular patches devoid of vegetation in the center but with perennial grasses at the margin. These patches occur in large numbers in the desert margin grasslands of southern Africa (Fig. 1, A and B). Early observers considered poisonous plants, ants, or termites as causal factors; however, most of these early hypotheses were systematically tested and rejected (1, 2). It has also been proposed that an unknown semi-volatile substance in the soil might be respon-

sible for the absence of grass within the FCs (2, 3). In fact, a wide range of volatile organic compounds are found in FCs (4). Measurements of carbon monoxide and hydrocarbons in the soil led to the proposal of a geochemical origin of FCs (5). Carnivorous ants (6) and “self-organizing vegetation dynamics” (7) have also been considered as causes for FCs. Despite the many hypotheses, the origin and the ecosystem function of FCs are still a much-debated mystery. I used a long-term data set describing the environmental and biogeographical characteristics and dynamics of FCs to identify the most likely cause of these unique formations. Additionally, I analyze the function of FCs in terms of

water management, biodiversity, and adaptation to arid conditions.

FCs occur along a narrow belt at the eastern margin of the Namib Desert, running from mid-Angola to northwestern South Africa. The area of distribution is closely associated with the isohyet of 100-mm mean annual precipitation (MAP) (Fig. 1B). The disjunct occurrence of FCs is caused by their pronounced restriction to sandy soils.

High soil humidity within FCs has been observed previously (1, 2). To confirm and quantify this potentially adaptive function, I measured volumetric soil water content ($\text{m}^3/\text{m}^3 \times 100$) from 2006 to 2012 within and around FCs. At sites with a MAP of 100 mm, more than 53 mm of water were stored in the upper 100 cm of soil, even during the driest time of the year (table S1). At a depth >40 cm, a soil humidity of more than 5% volumetric water content was recorded over 4 years.

Higher temporal resolution of water flux was gained by automatic measurements recorded every hour within the bare patch and the grass matrix at 10-, 30-, 60-, and 90-cm depths using FDR sensors. During the observation period of 4 years, the humidity at 60-cm depth within the FC was either at or well above 5% volumetric water content (Fig. 2A). In the typical sand texture of FC soils with dominant grain sizes around

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Forest Fungi

Boreal forest is one of the world's major biomes, dominating the subarctic northern latitudes of Europe, Asia, and America. The soils of boreal forest function as a net sink in the global carbon cycle and, hitherto, it has been thought that organic matter in this sink primarily accumulates in the form of plant remains. **Clemmensen et al.** (p. 1615; see the Perspective by **Treseder and Holden**) now show that most of the stored carbon in boreal forested islands in Sweden is in fact derived from mycorrhizal mycelium rather than from plant litter. Biochemical and sequencing studies show that carbon sequestration is regulated by functional and phylogenetic shifts in the mycorrhizal fungal community. The results will need to be explicitly considered in models of the role of the boreal forest in the global carbon cycle.

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